S SHRIRAM

Chennai, India | +91 9345598380 | shriram0981234@gmail.com | LinkedIn | GitHub | Portfolio | LeetCode

SUMMARY

Aspiring software engineer who builds AI-powered products end to end, from the models and algorithms to the application people actually use. Has built working systems for medical image screening, real-time fleet routing and post-quantum security, each taken from an idea to a usable app. Aims to grow into an engineer who designs reliable, well-built software and turns hard real-world problems into products people can depend on.

TECHNICAL SKILLS

Languages: Python, Java, SQL
AI / ML: PyTorch, timm, OpenCV, CNNs, Vision Transformers, Grad-CAM, NLP, RAG
Web & Backend: FastAPI, Flask, React, Vite, REST APIs
Data & Cloud: PostgreSQL, SQLite, MySQL, MongoDB, AWS (EC2, API Gateway, Amplify)
Tools: Git, GitHub, LangGraph, LangChain

PROJECTS

AuraMed – Women's Health Screening Platform | *React, FastAPI, PyTorch, timm, PostgreSQL* GitHub · Live · Video

- Built **AI-based detection** of breast cancer, cervical cell abnormalities and PCOS from medical images (ResNet-50, ViT-B/16), with **Grad-CAM heatmaps** explaining each result; achieved **95% accuracy** on PCOS (3,800+ scans) and **93.13%** on cervical cytology.
- Combined model output with clinical scoring (Gail model, Rotterdam criteria) into an **explainable risk score**, and built **portals for admins, doctors and patients**, where doctors digitally sign, lock, amend and release each report.

Hash Patch – Offline Post-Quantum Crypto Scanner | *Python, Tree-sitter, CycloneDX, React, Electron* GitHub · Installer · Video

- Built a scanner that parses Java, Python, Go and C code with Tree-sitter and traces **call graphs** to flag only the **cryptography that actually executes**, ignoring unused and test code.
- Designed it to export a signed **CycloneDX 1.6 Cryptographic Bill of Materials** and a **migration plan to post-quantum algorithms** (ML-KEM, ML-DSA), prioritised using Mosca's theorem, and packaged it as an **offline desktop app** for air-gapped systems.

FSOC-PAT – AI Beacon Tracking for Optical Communication | *Python, PyTorch, OpenCV, FastAPI, Electron* GitHub · Video

- Built a simulator for aligning free-space optical communication terminals (ISRO problem statement) that detects a moving beacon using **classical computer vision, a PyTorch CNN or a hybrid**, tracked with **Kalman/IMM filters** and a PID-controlled camera.
- Compared all three detection modes on identical trajectories and disturbances, exporting CSV and PDF reports with **measured pass/fail results**, and packaged the simulator as a Windows desktop app.

OptiFlow – Dynamic Route Optimisation | *Python, FastAPI, React, MapLibre GL, OSRM* GitHub · Video

- Built a **real-time fleet routing engine** that groups deliveries by location, schedules the most urgent stops first and optimises stop order (**2-opt**), while respecting hub deadlines, fuel capacity and each driver's 7.5-hour driving limit on real road networks.
- Simulated disruptions such as traffic, heavy rain, natural calamities and roadblocks that trigger **automatic re-routing**, with the reason for each change shown in separate dispatcher and driver views.

More projects across machine learning, full-stack web and desktop apps, with source code and demos, are available on my portfolio.

ACHIEVEMENTS

- **1st Place** – CUMTA Urban Mobility Challenge, Chennai (Nov 2026): inferred Metro-to-bus transfers from anonymised tap data
- **2nd Place** – AI Ascend 2026, Saveetha Engineering College: national hackathon backed by Kyndryl and AWS, with AuraMed
- **Theme Winner** (Transportation & Logistics) – CMR Hackfest 3.0 (Jan 2026): 36-hour national hackathon, with OptiFlow
- **1st Place** – Code Crux (Debugging & DSA), NIRAL '26, CEG, Anna University
- **1st Place** – Code Noobies, Carte Blanche '25, MIT Campus, Anna University
- **2nd Place** – Project Competition, NeXus 7.0 Industry Institution Summit, Easwari Engineering College (Oct 2025)
- **2nd Place** – Code Mathrix and Pseudo 2 Code, Mathrix '25, CEG, Anna University
- **3rd Place** – Reverse Coding, Tech Fiesta '25, Chennai Institute of Technology
- 500+ problems solved on LeetCode
- CBSE Class X school topper (97.4%); Class XII school rank 3 (95.8%)

CERTIFICATIONS

- **Oracle Certified Professional: Java SE 17 Developer** (1Z0-829), Mar 2026 · Verify
- **AWS Certified:** Solutions Architect – Associate, Developer – Associate, AI Practitioner, Cloud Practitioner (Jul 2026) · Verify
- **NPTEL (IITs):** Management Information Systems and Industry 4.0 & IIoT (Elite + Silver); Blockchain, NLP, Human-Computer Interfaces, Problem Solving in C (Elite); Discipline Star – CSE · Verify
- **Cisco:** Introduction to Networks (CCNA), Introduction to Modern AI, Security and Connectivity Support · Verify
- **1M1B × IBM SkillsBuild (AICTE):** AI for Sustainability Virtual Internship (Dec 2025 – Jan 2026) · Verify

EDUCATION

Easwari Engineering College, Chennai — **2024 – 2028**
B.E. Computer Science and Engineering — CGPA: 8.87

Indian Institute of Technology Madras (online, parallel degree) — **2024 – 2028**
BS in Data Science and Applications — CGPA: 7.52

Completed the Foundation level and the Diploma in Programming; Diploma in Data Science ongoing